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A clinical diagnosis of diurnal (non-sleep) bruxism in denture wearers

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SUMMARY **The purpose of this study was to establish a clinical method for diagnosing diurnal bruxism in denture wearers by recording masseter and anterior temporal electromygraph (EMG) activity. Seven suspected bruxists and five normal patients who wore complete dentures and/or distal extension base removable partial dentures were selected for participation. EMG activity in both the masseter and the anterior temporal muscles was recorded bilaterally during silent reading (10 min), maximal voluntary clenching (MVC), tapping in centric occlusion, lateral movements, chewing and swallowing. No significant differences of EMG activity were found between the groups during tapping, lateral movement, chewing and swallowing ($P > 0{\cdot}05$). However, during 10 min of silent reading, a significant difference was found between the groups when comparing masseter muscle activity ($P < 0{\cdot}05$). A threshold of 10% of MVC of at least 3-s duration was used to define an individual bruxism event. When the muscle activity recorded during silent reading was further analysed using these criteria, the control group displayed no bruxing activity while the suspected bruxist group displayed a mean frequency of six bruxism events (range 2–10). It was concluded that: (a) masseter muscle activity recorded during 10 min of silent reading showed significant difference between the groups; (b) the criteria selected in this study for the detection of sleep bruxism can also be used to assess diurnal bruxism; and (c) it is possible to diagnose diurnal bruxism in denture wearers by measuring the masseter EMG activity during 10 min of silent reading.**

Introduction

Several studies have attempted to establish the relationship between bruxism and temporomandibular disorders (TMD) (Olkinuora, 1969; Harold, 1970; Glaros & Rao, 1977; Kampe *et al.*, 1997; Lobbezoo & Lavigne, 1997) because bruxism is considered to be one of the most significant parafunction activities of the stomatognathic system.

Bruxism has been defined as the habitual grinding, clenching (Nadler, 1970; Rutrick, 1970; Glaros & Rao, 1977; Bowley, Stockstill & Pierre, 1993; Lavigne, Rompre & Montplaisir, 1996; Kampe *et al.*, 1997), gritting, clicking and gnashing of the teeth (Hudzinski & Walters, 1987), during the day and/or night for non-functional purposes. It occurs in practically all age groups and can be a habit carried throughout life, that is to say prior to the eruption of teeth, during the natural dentition period and through the artificial denture period (Nadler, 1970). According to Olkinuora's theory (Rao & Glaros, 1979), 'non-strain' bruxists are individuals who grind rather than clench and brux nocturnally, whereas 'strain' bruxists are daytime clenchers whose habit is a response to stress (Glaros, 1981).

Rugh & Solberg (1975) were the first to report the use of a portable electromyographic (EMG) device for monitoring muscle activity in bruxists. This enabled the measurement of bruxing activity in the subject's environment and it helped to clarify nocturnal bruxism behaviour under natural conditions (Clark, Beemsterboer & Rugh, 1981; Clarke & Townsend, 1984; Bowley *et al.*, 1993; Ikeda *et al.*, 1996; Lavigne *et al.*, 1996).

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As is mentioned above, diurnal bruxism is mainly considered as tooth clenching, and is frequently associated with nervous tension or physical effort. When the teeth are clenched frequently or for long periods, various deleterious effects can occur, such as facial pain, abnormal tooth wear, periodontal pain, increased tooth mobility, muscle tenderness to palpation, increased muscle tonus and hypertrophy, and temporomandibular joint (TMJ) discomfort (Nadler, 1970; Rutrick, 1970; Glaros & Rao, 1977). Such problematic effects are directly related to both the intensity and the persistence of the abnormal activity (Wojnilower & Gross, 1981).

It can be difficult to establish whether a patient is bruxing or not because, in many cases, neither the patient nor the dentist are aware of the habit and clear signs and/or symptoms are not always present. Thus, diagnosis of bruxism can be difficult and numerous techniques and criteria are currently in use (Ikeda *et al.*, 1996; Lavigne *et al.*, 1996). Despite the variations in the criteria used to either diagnose or to assess bruxism behaviour, an accurate method has not been developed that can recognize a diurnal bruxist from a normal subject. In other words, all of the criteria currently in use are related to nocturnal or sleep-associated bruxism. In clinical practice, it is not uncommon to deal with denture wearing patients who frequently experience pain in the underlying soft tissue, even though the denture is been considered satisfactory (retention, adaptation, stabilization, occlusal balance, etc). Because bruxism occurs in practically all age groups, including during the artificial denture period, we hypothesize that many of the pain-related complaints made by these patients could be the result of a clenching or tapping form of diurnal bruxism.

Thus, the purpose of this study was to establish a clinical method for diagnosing diurnal bruxism in denture wearers by using recordings of the masseter and the anterior temporal EMG activity during the performance of different tasks.

Materials and methods

Subjects

Twelve adults were selected from the patient population of the Tokyo Dental College for participation in the study. The gender distribution was three females and four males in the suspected bruxist group and two females and three males in the control group (mean age 67·6 years; range 51–82 years). All of the subjects were fully informed and agreed to participate in the study. They were either completely or partially edentulous patients (distal extension base removable partial denture) who had been using the dentures for several months (at least 2 months) following the final adjustments; all of the appliances had satisfactory retention, adaptation, stabilization and occlusal balance. The seven subjects selected as suspected bruxists had complained of pain in the underlying soft tissue, especially in the afternoon. Five subjects who did not have a past history of bruxism and who were not experiencing pain in the underlying soft tissue were selected as controls for comparative purposes.

All of the subjects underwent an identical clinical examination based on a questionnaire to assess past history of bruxism (awareness) and the presence of pain or discomfort, and on an examination of the masticatory and neck–shoulder muscles and of the TMJ (joint noises were registered during movements of the mandible). The dentures were also assessed with regard to retention, adaptation, stabilization, occlusion and vertical dimension.

Recording system

An EMG recording device* was used to record the myoelectrical activity of the masseter and anterior temporal muscles. By means of electrodes applied directly to the skin, the device measured the EMG from four muscles simultaneously. To maximize the signal-to-noise ratio, a differential amplifier was connected directly to the reference (grounding) electrode, thus eliminating disturbances such as those caused by movements.

The EMG preamplifier measurement sensitivity was $\pm 1\ \mu V$. The measurement range for bipolar EMG signals was $\pm 5000\ \mu V$. Measurement data were downloaded into the computer using an interface unit. In this way, the measurement values of each channel were displayed and analysed using a software program.

The sampling frequency selected in the study was 1000 Hz at a sampling period of 0·1 s. Four channels were used to record both left and right masseter and anterior temporal muscles.

* Muscle Tester ME3000p Professional System, Mega Electronics Ltd., Kuopio, Finland.

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During the pre-experimental phase of this study, the accuracy of the device was evaluated by recording EMG activity while moving the head and neck in different directions and by observing the possible influence of electromagnetic waves (TV, telephone and fluorescent lights) on the EMG recordings.

Experimental procedure

For EMG registration of muscle activity, bipolar surface electrodes† were positioned parallel to the main direction of the muscle fibres, on the maximal bulk of the masseter and the anterior temporal muscles, which was determined by palpation while the subjects clenched intermittently. The interelectrode distance was 2·5 cm centre-to-centre. The reference (grounding) electrodes with the integrated preamplifier were attached behind the ear. The skin impedance between the two signal electrodes was measured during the set-up phase. Prior to the attachment of the electrodes, the skin was thoroughly cleansed with a specific Skin Cleansing Gel‡ used for bioelectrical measurements.

During the recording procedures, each subject was comfortably seated upright in a dental chair with the head unsupported, and was asked to carry out the following tasks:

(a) Silent reading for 10 min: the subjects were asked to read in silence some printed material from a book, newspaper or magazine. No further instructions were given. Muscle activity during this task was recorded.
(b) Tapping contacts in centric occlusion: the subjects were instructed to tap in centric occlusion 10 times, with a duration of approximately 1 s for each tap.
(c) Lateral movements: the subjects were asked to move their mandible from right to left five times.
(d) Maximal voluntary clenching (MVC): the subjects were asked to clench as strong as possible three times with a duration of approximately 8 s for each clench. Between clenches, the subjects were instructed to relax for about 30 s in order to avoid muscle fatigue.
(e) Chewing: each subject was instructed to chew one soft candy and then swallow it.
(f) Swallowing: muscle activity during the drinking of 10 mL of water was recorded. This activity was carried out twice.

Data analysis

In order to compare the results from muscles on different sides and different subjects, the EMG results were expressed as a percentage of MVC. The Mann–Whitney U-test was used for statistics analysis. We selected $P < 0{\cdot}05$ as the statistically significant level.

For MVC clench, the mean voltage amplitude of each clench was measured. Each muscle's maximal value was then selected.

The muscle activity measured during silent reading was sectioned at 1-min intervals, and the mean voltage amplitude of each interval was measured. The average of 10 measurements was then calculated.

For tapping in centric occlusion, the mean voltage amplitude of each tap was measured. Muscle activity was based on the average of 10 taps.

For mandibular lateral movements, the mean voltage amplitude of each movement was measured. Muscle activity was based on the average of five lateral movements.

The muscle activity during chewing was calculated by measuring the mean voltage amplitude of the whole activity, from the moment the patients began to chew the soft candy until the moment they swallowed it.

For swallowing measurements, the mean voltage amplitude of each swallow was measured; muscle activity was based on the average of two swallows.

Once the individual results were obtained, statistical tests were carried out in order to compare both groups' results. The hypothesis examined in this study was that significant differences in muscle activity could be found between control and suspected bruxists group during silent reading. The mandibular position was not determined during the silent reading exercise but, because no further instructions were given to the subjects, it was postulated that it would be relaxed. If the results were as we expected, it was thought that further analysis would be necessary in order to determine whether or not bruxism events were present.

† Disposable prefilled Ag/AgCl electrodes. Blue Sensor. Medicotest, Ølstykke, Denmark.
‡ Skin Pure, Nihon Kohden, Tokyo, Japan.

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Table 1. Demographic features of control and suspected bruxist group

	Control	Suspected bruxist
Number of subjects	5	7
Female/male	2/3	3/4
Mean age (years)	73	61
Denture status		
Upper/lower: complete denture	4	4
Upper/lower: partial denture	—	1
Upper: complete denture; lower: partial denture	1	—
Upper: partial denture; lower: complete denture	—	1
Upper: complete denture; lower: natural teeth	—	1
Clinical findings		
Pain in the underlying soft tissue during the afternoon	—	7
Muscle tenderness	2	2
Joint sound	—	1
Mandible deviation	—	—
Headache	—	2
Stiff neck	—	—
Bruxism awareness	—	4

The criteria selected to assess the presence of bruxism events were based on a study carried out by Ikeda *et al.* (1996). A bruxism event was defined as the minimum level of 10% of MVC of at least 3-s duration. With regard to the linkage criterion, a minimum of 4-s duration between EMG elevations was selected before they were considered to be separate events.

Results

The demographic features of both groups are presented in Table 1.

The MVC values of both the control and the suspected bruxist groups for the masseter and the anterior temporal muscles are shown in Table 2. As was previously mentioned, the mean value of each EMG activity is expressed as %MVC.

EMG measurements revealed that there were no significant differences between the controls and the suspected bruxists during tapping in centric occlusion, mandibular lateral movements, chewing and swallowing (see Table 3). However, the results obtained when the muscle activity was recorded during silent reading (10 min) showed that the suspected bruxist group experienced significantly more masseter muscle activity than did the control group (see Table 3 and Figs 1 and 2 for data and statistics). The temporal muscle did not display any statistical significant difference between groups because even the control group exhibited a constant activity in the temporal muscle with few periods of electrical silence.

Because a statistically significant difference was found between groups when muscle activity during silent reading was analysed, we decided to carry out further analyses. In order to find out whether or not the observed muscle activity was the result of bruxism events, specific criteria for the detection of bruxism were applied. As was previously mentioned, the criteria were based on the study carried out by Ikeda *et al.* (1996) in which 10% of a MVC minimum threshold of at least 3-s duration was used to define an individual

Table 2. Maximal voluntary clenching values for both groups

Subjects	Right masseter[a] (μV)	Left masseter[a] (μV)	Right temporal[a] (μV)	Left temporal[a] (μV)
C1	65 ± 25	78 ± 27	26 ± 13	11 ± 6
C2	25 ± 12	30 ± 11	35 ± 12	33 ± 12
C3	87 ± 32	70 ± 26	42 ± 21	35 ± 13
C4	23 ± 10	25 ± 11	64 ± 23	59 ± 20
C5	74 ± 22	85 ± 22	54 ± 19	22 ± 10
B1	56 ± 25	97 ± 33	87 ± 34	70 ± 32
B2	55 ± 16	49 ± 15	32 ± 9	37 ± 11
B3	81 ± 18	109 ± 25	49 ± 11	53 ± 14
B4	40 ± 13	36 ± 11	55 ± 19	48 ± 16
B5	150 ± 43	125 ± 41	215 ± 65	219 ± 42
B6	43 ± 21	51 ± 34	30 ± 20	39 ± 21
B7	211 ± 58	117 ± 65	133 ± 35	138 ± 45

[a] Mean levels ± s.d. of the voltage amplitude.
C = control group; B = suspected bruxists group.

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Table 3. Electromyographic results

Tasks	Muscles	Control $n = 5$	Suspected bruxists $n = 7$	Statistics[a]
Tapping in central occlusion[b]	Masseters	22 ± 16·9 (7–43)	38 ± 28·9 (10–92)	0·37
	Temporals	27 ± 8·8 (17–41)	30 ± 22·3 (8–72)	0·93
Lateral movements[b]	Masseters	23 ± 17·8 (2–50)	24 ± 19·0 (5–60)	0·87
	Temporals	26 ± 15·5 (6–47)	22 ± 11·8 (13–46)	0·37
Chewing a soft candy[b]	Masseters	49 ± 21·5 (28–81)	59 ± 11·6 (48–78)	0·37
	Temporals	57 ± 20·1 (42–92)	50 ± 11·3 (33–64)	0·62
Swallowing of water[b]	Masseters	17 ± 8·5 (7–30)	33 ± 26·7 (6–84)	0·32
	Temporals	27 ± 10·3 (12–38)	27 ± 19·0 (6–24)	0·62
Silent reading[b]	Masseters	0 ± 0·44 (0–1)	5 ± 3·6 (1–9)	0·014*
	Temporals	3 ± 3·2 (0–8)	6 ± 4·0 (1–13)	0·32

[a] P value = Mann–Whitney U-test.
[b] Mean ± s.d., with range in parentheses. Values are expressed as a percentage of MVC.
* $P < 0·05$.
Muscle activity is expressed as a percentage of MVC.

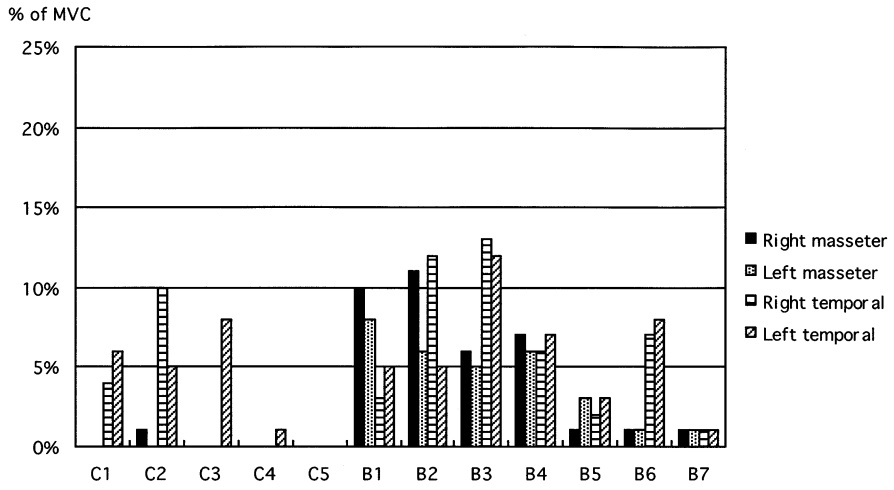

Fig. 1. Individual EMG muscle activity during silent reading.

bruxism event. With regard to the linkage criterion, the literature records a variety of duration, ranging from 2 to 10 s (Wagner, 1981; Piccione *et al.*, 1982; Kato *et al.*, 1989; Ikeda *et al.*, 1996). The present study selected a 4-s linkage criterion.

Thus, bruxism events (based on the above criteria) were not observed in the control group. The anterior temporal muscles displayed a constant EMG activity but this did not concur with the activity of the masseter muscle. In the suspected bruxist group, the masseter muscles displayed more EMG activity than did the control group. Various distinct patterns of muscle activ-

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ity were recognized. The following results were obtained when the criteria for the detection of bruxism events were applied.

Total bruxism events

During the 10 min of silent reading, there was a total of 43 events for all subjects from the suspected bruxist group. The mean frequency of bruxism events was six (range 2–10 events).

Duration of bruxism events

The average duration for all bruxism events, as well as the total duration of bruxism activity, were calculated individually. The total duration of bruxism activity ranged from 21·8 to 260·1 s.

EMG level per event

The EMG level per bruxism event (mean value) was also calculated individually. The values ranged from 11 to 41% MVC.

The results are shown in Table 4 and Fig. 3.

Discussion

The reasons behind the lack of studies regarding diurnal bruxism are not well understood. It may be because of the difficulty of recording muscle activity under natural conditions, the fact that at times neither the patient nor the dentist are aware of the presence of bruxism, or because of its day-to-day variable nature.

In dental practice, especially in the case of denture wearers, it is sometimes difficult to clarify the origin of a complaint because there are many possible influencing factors related to them, such as the condition of the dentures, occlusion state, age, gender, medical conditions. It is relatively easy for the clinician to critically assess most of these factors through historical and clinical observation. However, the presence of parafunctions in denture wearers as a contributing factor to the discomfort of the dentures has been less understood, generally because of a lack of clinical methods to critically assess them. In this study, it is suggested that a considerable number of complaints relating to pain in the soft tissue could be the result of sustained pressure over the underlying tissue caused by a diurnal bruxism habit. The experiments were, therefore, designed to attempt to establish a clinical method for the diagnosis of diurnal bruxism in denture wearers.

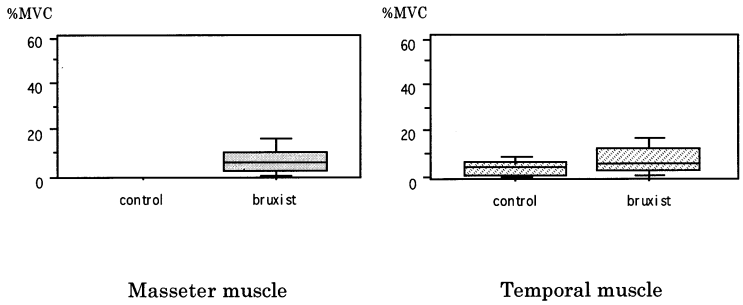

Fig. 2. Comparison of EMG activity between groups during silent reading (Mann–Whitney *U*-test, $P \leqq 0{\cdot}05$ significant difference). For masseter muscle, the *P*-value = 0·014 (significant difference) and for temporal muscle the *P*-value = 0·329.

Table 4. Results obtained when bruxism events detection criteria were applied in a suspected bruxist group

Variables	B1	B2	B3	B4	B5	B6	B7
Frequency of bruxism events	4	9	6	10	9	2	3
Duration per event[a] (s)	65·0 ± 32·0 (36·9–110·6)	20·3 ± 20·7 (3·4–55·6)	32·7 ± 26·7 (5·8–80·6)	7·1 ± 6·4 (3·5–24·2)	5·3 ± 1·7 (3·7–8·7)	17·8 ± 16·8 (5·9–29·7)	7·3 ± 1·8 (6·2–9·4)
Total duration of bruxing activity (s)	260·1	182·8	196·7	71·2	47·5	35·6	21·8
EMG level per event[b] (%MVC)	12 ± 3	19 ± 7·3	15 ± 3·5	41 ± 9·3	15 ± 4·4	13 ± 7·8	11 ± 1·6

[a] Mean ± s.d. with range in parentheses.
[b] Mean ± s.d.

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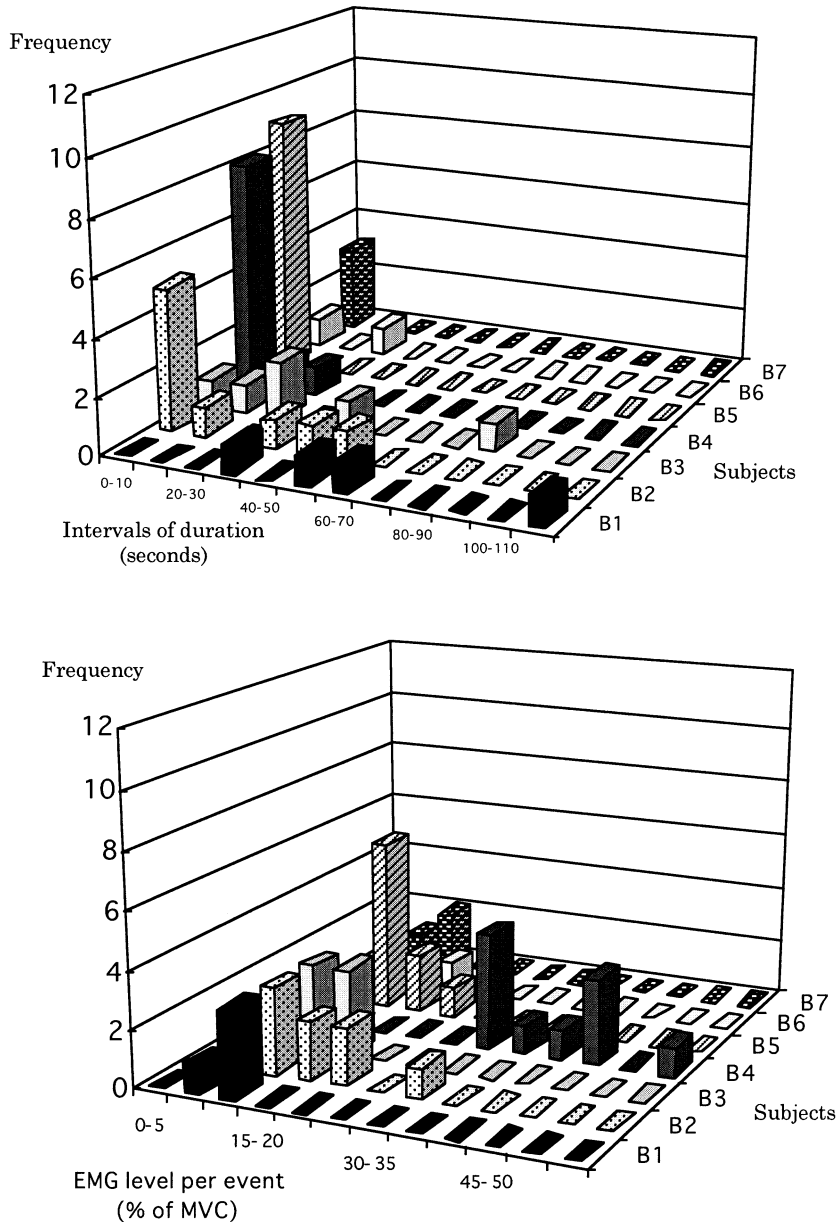

Fig. 3. Individual frequency, duration and EMG level of each bruxism event observed during silent reading.

In this study, the assumed hypothesis was that bruxists display higher muscle activity than do normal subjects (Rutrick, 1970; Hudzinski & Walters, 1987; Kampe *et al.*, 1997), especially during rest or in an unstressed position. Sheikholeslam, Moller & Lous (1982) reported that patients with oro-facial pain and functional disorders show higher postural muscle activity than do normal subjects, when measured by EMG.

EMG recordings of masticatory muscles have been used for many years as a method for studying their activity during normal functions, e.g. chewing, swallowing, yawning, speaking, etc., and during parafunctions, e.g. sleep bruxism (Rugh & Solberg, 1975; Solberg, Clark & Rugh, 1975; Clark *et al.*, 1981; Clarke, Townsend & Carey, 1984; Clarke & Townsend, 1984; Ikeda *et al.*, 1996; Lavigne *et al.*, 1996).

Based on the statement that bruxists displayed higher EMG levels than did normal subjects (Rutrick, 1970; Sheikholeslam *et al.*, 1982; Hudzinski & Walters, 1987; Kampe *et al.*, 1997), muscle activity during different situations (tapping, lateral movement, chewing, swallowing, silent reading, etc.) was recorded and

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compared. Contrary to what might have been expected, no significant differences in the levels of masseter and temporal muscle activity were observed between the groups. There were individual variations and the suspected bruxist group displayed slightly higher levels than did the normal group, but statistical tests showed no significant differences (Table 3). However, the suspected bruxist group displayed a markedly higher masseter muscle activity during silent reading (which was considered an unstressed position) than did the normal subjects (Table 3 and Figs 1 and 2). The reason for these results may be because subjects from both groups, when asked to carry out the different tasks, became too conscious of their actions. However, during the silent reading task, no specific instructions were given; thus, the subjects were free to perform an oral parafunction subconsciously. To verify the possibility of day-to-day variations, two sessions of recordings were carried out for both groups. The same results were found in each.

From the present study, it was apparent that the temporal muscles in both groups displayed a constant activity during silent reading; when the results were compared, no significant difference was found between the groups. These findings concur with those of others (Tallgren *et al.*, 1980; Majewski & Gale, 1984) and suggest that the temporal muscle is markedly sensitive to changes in jaw and occlusal positions, giving rise to considerable individual variation. Moreover, because the temporal muscles are responsible for maintaining the stability of the mandible, their activity is dependent on jaw position. During silent reading, the position of the head was not fixed because it was thought that this could produce a stress factor that could influence the recordings under natural condition. Thus, during silent reading, it is possible that the mandible displaces either forward or laterally, displaying a constant EMG activity. Other recording methods, such as a mandibular movement recording device, were not used because not only could they be an obstacle to achieving an unstressed or physiological relaxed position, but also because they do not reveal the level of muscle activity during different situations.

Even though significant difference was found between the groups when the results of masseter muscle activity during silent reading were compared, it remained unclear whether bruxism events were present. In order ensure that the results obtained were caused by the presence of diurnal bruxism activity, the EMGs were further analysed using a bruxism detection criteria.

There are numerous cut-off criteria for the detection of bruxism, such as those established by Rugh & Solberg (1975), Clarke *et al.* (1984), Lobbezoo & Lavigne (1997) and Ikeda *et al.* (1996). The criteria suggested in the study by Ikeda *et al.*, although developed to assess sleep bruxism, were selected for this study because we considered them to be complete criteria, including data regarding changes in the electrocardiogram and the electroencephalogram while bruxism events were presented, and also because they expressed muscle activity as a percentage of MVC. This was thought to be more accurate than selecting an arbitrary threshold, ignoring the subject's maximal clenching force. Many studies have stated that nocturnal and diurnal bruxism have different aetiological bases (Olkinuora, 1969; Glaros, 1981; Kampe *et al.*, 1997) and should be assessed as a separate behaviour. However, we considered that the muscle activity *per se* does not display great differences between the diurnal and the nocturnal type.

When the bruxism detection criteria were applied, it was possible to effectively demonstrate that a subject from the suspected bruxist group displayed bruxism events, while no events were observed in the control group, except for some activity of low intensity possibly related to swallowing. The results showed clearly recognizable different individual patterns of diurnal bruxism activity (Fig. 3): (a) high voltage amplitude, short duration but high frequency (subject B4); (b) low voltage amplitude levels, long lasting at each event, low frequency (subject B1); (c) high frequency, medium voltage amplitude levels and long lasting (subject B2 and B3); (d) medium voltage amplitude levels, high frequency but short duration (subject B5); and (e) medium voltage amplitude levels, low duration and low frequency (subject B6).

To date, only two basic patterns of bruxism activity have been defined: a rhythmic chewing like movement and a prolonged strong isotonic contraction (Ikeda *et al.*, 1996; Lavigne *et al.*, 1996). These take two variables into consideration: voltage amplitude levels and duration of the activity. In the literature, there are many words defining the different patterns of bruxism, such as gnashing, clenching, grinding, bracing, tapping, etc. However, it is known that, through the observation of EMGs, it is difficult to distinguish between a

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grinding and a clenching event. For this reason, we considered that more important than selecting a name for a specific pattern was to have a clear understanding of the role of three factors: duration, intensity and frequency, all of which are necessary for defining the pathology of bruxism.

The hypothesis that complaints made by denture wearers were the result of daytime bruxism was supported, because no other reasons for their discomfort were presented at oral examination. It has been reported that pressure can cause histopathological changes in the supporting tissues (Hara *et al.*, 1996; Mori *et al.*, 1997). These changes (bone resorption, epithelial tissue and lamina propria mucosa compression, inflammatory changes, deformation and shortening of the epithelial ridges) are correlated to the strength and the duration of the continuous pressure. Thus, as a result of pressure exerted by prolonged, forceful and/or frequent bruxism, the mucosa underlying the denture can become painful.

Parafunctions such as bruxism (nocturnal or diurnal) should be identified prior to the design of any treatment plan. This is especially important in the case of denture wearers, where the clinical diagnosis of diurnal bruxism is a key for the success of their dental treatment.

Acknowledgments

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